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Self-Consistent Kinetic Monte Carlo Simulation of Deep Dielectric Charging in Spacecraft Insulation

Application to a Simplified Multilayer Satellite Solar Panel

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Abstract

Spacecraft operating in the near-Earth space environment are continuously exposed to energetic charged particles that interact with spacecraft materials and may lead to the accumulation of electrical charge within dielectric components. Among the various charging phenomena, deep dielectric charging is of particular concern because energetic electrons can penetrate insulating materials and deposit charge beneath the surface, generating internal electric fields that may eventually result in electrostatic discharge and degradation of spacecraft subsystems.

The objective of this internship was to develop a computational framework for investigating deep dielectric charging in a multilayer spacecraft solar panel using the Geant4 Monte Carlo simulation toolkit. A simplified multilayer geometry consisting of a SiO_2 cover glass, silicone adhesive, silicon solar cell, Kapton insulation and aluminium substrate was implemented to represent the principal structural layers of a spacecraft solar panel. Monoenergetic electron beams covering energies from 1 keV to 5 MeV were simulated to study electron transport through the multilayer structure.

The simulation outputs were analysed using custom ROOT-based post-processing routines to evaluate secondary particle production, primary interaction statistics, total and layer-wise energy deposition, depth-dose profiles, charge accumulation, charge density, electric field and potential difference. These analyses establish a systematic workflow linking microscopic particle interactions to the macroscopic electrostatic response of the multilayer structure.

The developed framework provides a flexible platform for investigating radiation-induced charging phenomena in spacecraft dielectric materials and may be extended in future studies to incorporate realistic space radiation environments, time-dependent charge transport models and self-consistent charge transport and electrostatic field solvers.

Introduction

Spacecraft operating in Earth orbit are continuously exposed to energetic charged particles originating from the solar wind, solar energetic particle events, and the Earth's radiation belts[9,13]. The interaction of these particles with spacecraft materials leads to a variety of radiation effects, including material degradation, single-event effects and spacecraft charging. Among these phenomena, spacecraft charging represents a significant reliability concern because the accumulation of electrical charge can produce electrostatic discharges capable of damaging onboard electronics and degrading spacecraft performance.

Deep dielectric charging[7,8] occurs when energetic electrons penetrate insulating materials and become trapped beneath the material surface. Unlike surface charging, where charge accumulates on externally exposed surfaces, deep dielectric charging produces internal electric fields within dielectric materials. If the accumulated charge exceeds the dielectric strength of the material, electrostatic discharge may occur, resulting in permanent damage to spacecraft components.

Spacecraft solar panels contain multiple dielectric and conductive layers that are continuously exposed to energetic electron environments throughout their operational lifetime. Consequently, understanding electron transport, energy deposition and charge accumulation within these multilayer structures is essential for improving spacecraft reliability and mitigating charging-related failures. Monte Carlo particle transport simulations[1,2] provide an effective approach for analysing these processes because they explicitly model the interaction of energetic particles with complex material geometries.

1.1 Objectives

The primary objective of this internship was to develop a Geant4-based computational framework for analysing deep dielectric charging in a multilayer spacecraft solar panel subjected to energetic electron irradiation. The specific objectives of the work are summarized as follows:

- Develop a multilayer spacecraft solar panel model using the Geant4 simulation toolkit.
- Simulate the transport of monoenergetic electrons through the multilayer structure.
- Analyse secondary particle production and primary interaction statistics.
- Evaluate total, layer-wise and depth-dependent energy deposition within the solar panel.
- Estimate charge accumulation, charge density, electric field and potential difference from the simulated electron transport data.
- Develop a modular simulation framework that may be extended for future investigations of spacecraft charging phenomena.

1.2 Scope of the Internship

The present work focuses on the development of a simulation framework for analysing deep dielectric charging in a simplified multilayer spacecraft solar panel. The study considers monoenergetic electron irradiation over an energy range of 1 keV to 5 MeV and investigates the resulting electron transport and electrostatic behaviour within the multilayer structure.

The scope of the internship is limited to single-exposure charge accumulation based on Monte Carlo particle transport simulations. Time-dependent charge transport, electrostatic discharge mechanisms, realistic space radiation spectra and spacecraft-wide charging effects are beyond the scope of the present work. Nevertheless, the developed framework provides a suitable foundation for incorporating these capabilities in future investigations.

Theoretical Background

This chapter presents the theoretical concepts relevant to deep dielectric charging in spacecraft solar panels. It introduces the near-Earth radiation environment, the mechanisms responsible for spacecraft charging, the structure of multilayer solar panels and the principal interactions governing energetic electron transport in matter.

2.1 Space Radiation Environment

Spacecraft operating in Earth orbit are continuously exposed to a complex radiation environment consisting primarily of electrons, protons and heavy ions originating from various natural sources. The principal contributors include the solar wind, solar energetic particle (SEP) events and the trapped particle populations of the Earth's Van Allen radiation belts.[9,13]

The intensity and energy of these particles depend strongly on the orbital altitude and mission profile. Low Earth Orbit (LEO) spacecraft primarily encounter trapped radiation while periodically passing through regions such as the South Atlantic Anomaly. In contrast, spacecraft operating in Geostationary Earth Orbit (GEO) are continuously immersed in energetic electron populations capable of penetrating spacecraft dielectric materials.

Energetic electrons with energies ranging from several keV to several MeV are of particular importance for spacecraft charging studies. These electrons penetrate insulating materials, deposit energy beneath the surface and gradually accumulate charge within dielectric layers. Understanding the interaction of these particles with spacecraft materials is therefore essential for evaluating radiation-induced charging effects.

2.2 Spacecraft Charging

Spacecraft charging[6,7,8] refers to the accumulation of electrical charge on or within spacecraft materials due to interactions with the surrounding plasma and energetic particle environment. Depending on the location of charge accumulation, spacecraft charging is generally classified into surface charging and deep dielectric charging.

Surface charging occurs when low-energy plasma particles accumulate on externally exposed spacecraft surfaces. Differential charging between adjacent materials may produce electrostatic discharge, potentially disturbing onboard electronics and communication systems.

Deep dielectric charging, on the other hand, is produced by energetic electrons capable of penetrating insulating materials. As these electrons lose energy through successive interactions, a fraction becomes trapped beneath the material surface, producing internal electric fields within the dielectric. Continued accumulation of charge may eventually exceed the dielectric strength of the material and initiate an internal electrostatic discharge.

Among these charging mechanisms, deep dielectric charging represents a significant reliability concern for spacecraft operating in energetic electron environments because the accumulated charge cannot dissipate rapidly through insulating materials.

2.3 Multilayer Spacecraft Solar Panels

Spacecraft solar panels consist of multiple dielectric and conductive layers that collectively provide structural support, environmental protection and electrical power generation. Although the exact configuration varies depending on spacecraft design, a typical solar panel comprises a protective cover glass, adhesive layers, photovoltaic cells, electrical insulation and a structural substrate.[10,11]

In the present work, a simplified multilayer geometry consisting of SiO_2 cover glass, silicone adhesive, silicon, Kapton insulation and aluminium substrate is considered. The SiO_2 cover glass protects the photovoltaic cell from ultraviolet radiation, atomic oxygen and micrometeoroid impacts. Silicone adhesive provides mechanical bonding between adjacent layers, while the silicon layer functions as the active photovoltaic material. Kapton serves as an electrical insulator and thermal protection layer, whereas the aluminium substrate provides structural rigidity.

The presence of multiple dielectric interfaces causes the transport and deposition of energetic electrons to vary significantly between individual layers, making multilayer solar panels particularly suitable for investigating deep dielectric charging phenomena.

2.4 Electron–Matter Interactions

As energetic electrons propagate through matter, they continuously interact with the atoms of the material through several electromagnetic processes.[10,11] These interactions determine the penetration depth of the incident electrons, the amount of deposited energy and the generation of secondary particles.

Ionization[10,11] represents the dominant mechanism of energy loss for electrons over the energy range considered in this work. During ionization, incident electrons transfer part of their kinetic energy to atomic electrons, producing electron–hole pairs and secondary electrons. Multiple Coulomb scattering continuously alters the electron trajectory through repeated elastic interactions with atomic nuclei, increasing the effective path length travelled within the material. At higher energies, electrons may also emit bremsstrahlung photons as they are decelerated within the electric field of atomic nuclei.

The combined effect of these interactions governs secondary particle production, and energy deposition within the multilayer structure. These quantities form the basis for evaluating charge accumulation and the resulting electrostatic behaviour investigated in the subsequent chapters.

Simulation Methodology

This chapter describes the simulation framework developed to investigate deep dielectric charging in a multilayer spacecraft solar panel. The framework combines Monte Carlo particle transport simulations performed using Geant4 with custom ROOT-based post-processing routines to analyse electron transport, energy deposition and the resulting electrostatic behaviour. The simulation geometry, material configuration, irradiation conditions and data analysis procedure adopted throughout this work are presented in the following sections.

3.1 Simulation Framework

The computational framework developed during this internship integrates the Geant4 Monte Carlo toolkit[1,2,4] with ROOT-based data analysis routines[5] to study deep dielectric charging in spacecraft dielectric materials. Geant4 is employed to simulate the transport of energetic electrons through a multilayer solar panel while accounting for the relevant electromagnetic interactions within each material. The resulting simulation data are subsequently processed using ROOT to extract quantities associated with particle transport, energy deposition and electrostatic charging.

The overall workflow begins with the definition of the simulation geometry, material properties and primary electron beam. Geant4 then performs Monte Carlo particle transport, recording quantities such as secondary particle production, interaction statistics and energy deposition. The resulting ROOT output is subsequently processed using dedicated analysis routines to evaluate charge accumulation, charge density, electric field and potential difference.

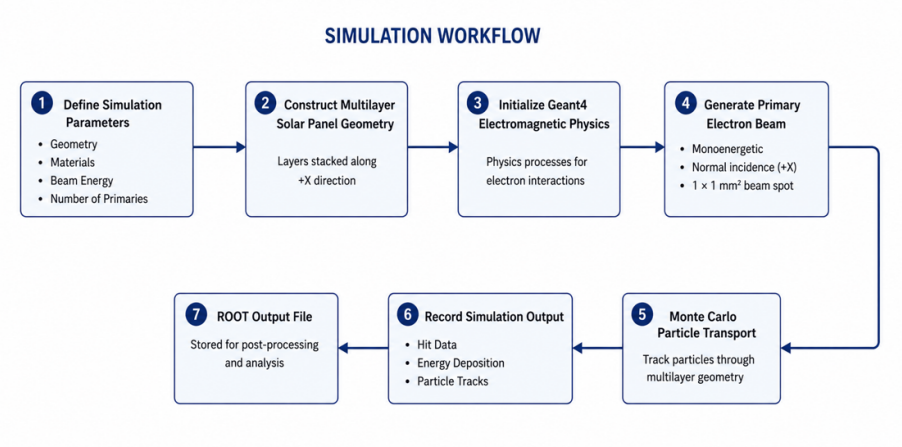


Figure 3.1: Overall simulation workflow illustrating the Geant4 particle transport stage and the subsequent ROOT-based post-processing analyses used to evaluate deep dielectric charging.

The modular design of the framework allows individual analyses to be performed independently while using the same simulation output. Consequently, additional materials, geometries or charging models may be incorporated without significant modifications to the underlying simulation architecture.[3]

3.2 Solar Panel Geometry

A simplified multilayer spacecraft solar panel was implemented in Geant4 using standard solid volumes. The adopted geometry represents the principal structural and functional layers commonly found in spacecraft solar array assemblies while remaining computationally efficient for Monte Carlo simulations.

The layers are stacked sequentially along the positive X-direction, with the incident electron beam entering normally through the SiO_2 cover glass. The multilayer configuration consists of a protective SiO_2 cover glass, silicone adhesive layers, a silicon solar cell, Kapton insulation and an aluminium substrate. This arrangement enables the investigation of electron transport across multiple dielectric and conductive interfaces, thereby providing a representative model for studying deep dielectric charging. No vacuum gaps were included between adjacent layers.

Table 3.1: Layer configuration of the simulated spacecraft solar panel.

Layer	Material	Thickness (μm)
1	SiO_2 (Cover Glass)	100
2	Silicone Adhesive	25
3	Silicon Solar Cell	180
4	Silicone Adhesive	25
5	Kapton	50
6	Aluminium	500
	Total Thickness	880

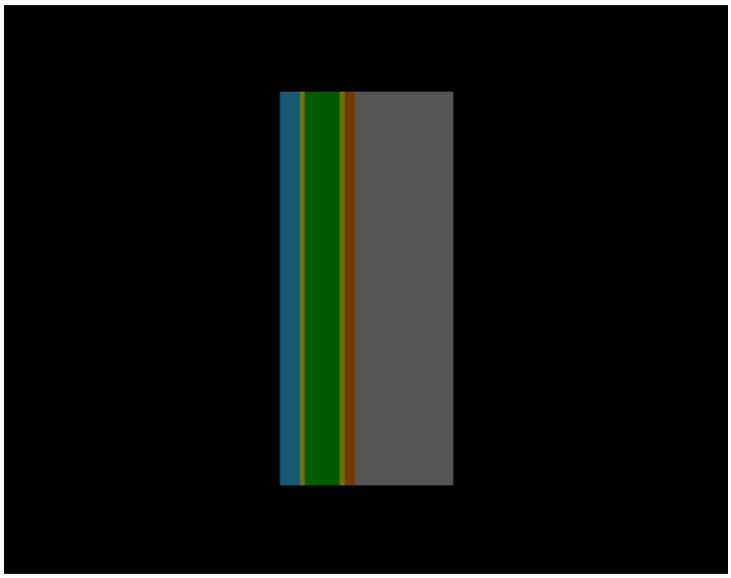


Figure 3.2: Simplified multilayer spacecraft solar panel geometry implemented in Geant4. The incident electron beam enters normal to the panel surface along the positive X-direction.

3.3 Simulation Parameters

The simulations were performed using monoenergetic electron beams incident normally on the multilayer solar panel. Identical geometry and physics configurations were maintained for all simulations to enable direct comparison between different incident energies.

Table 3.2: Simulation parameters used throughout the study.

Parameter	Value
Primary particle	Electron
Number of primaries	2000 per simulation
Beam dimensions	1×1 mm^2
Beam direction	+X
Beam incidence	Normal to panel surface
Source distance	1.5 mm
No. of incident energies	15
Incident energy range	1 keV – 5 MeV
Geometry dimensions	2×2 mm^2
Simulation toolkit	Geant4 11.3.0
Analysis software	ROOT 6.36

The selected energy range spans from low-energy electrons that stop within the upper dielectric layers to high-energy electrons capable of penetrating the complete multilayer structure. This energy sweep enables the investigation of electron transport and charging behaviour over a wide range of irradiation conditions representative of the near-Earth space environment.

3.4 Post-Processing Workflow

Although Geant4 records detailed information regarding particle transport and energy deposition, several quantities relevant to deep dielectric charging are obtained through post-processing of the simulation output. A dedicated analysis framework was therefore developed using CERN ROOT to process the generated ROOT files and extract engineering quantities associated with spacecraft charging.

The post-processing routines analyse the event-level simulation data to determine secondary particle production, primary interaction statistics, total and layer-wise energy deposition, depth-dose profiles, charge accumulation, charge density, electric field and potential difference. Each analysis module operates independently while using a common simulation output, allowing new analyses to be incorporated without modifying the Geant4 simulation itself.

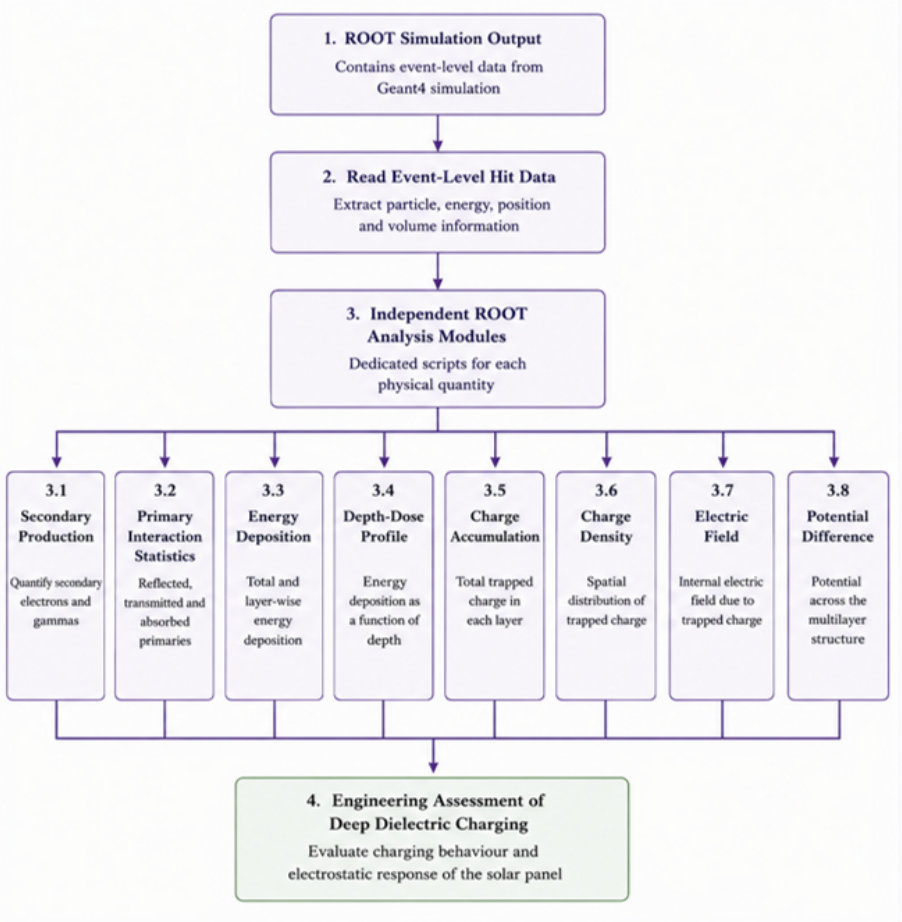


Figure 3.3: ROOT-based post-processing workflow showing the independent analysis modules developed to extract radiation transport and electrostatic charging quantities from the Geant4 simulation output.

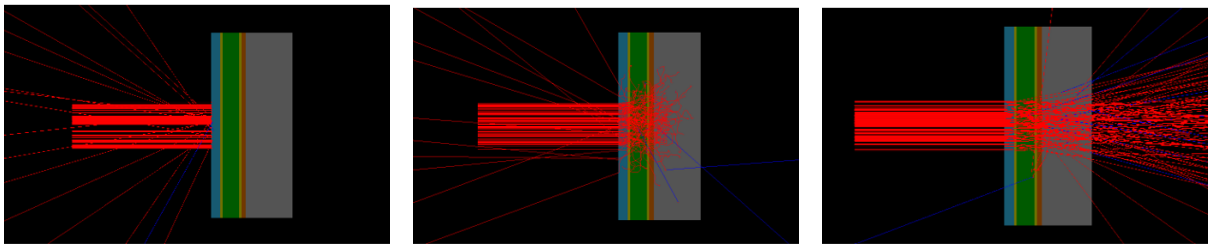
The separation of particle transport and data analysis provides a flexible and extensible simulation framework suitable for future investigations involving additional material systems, irradiation environments and charging models.

Results and Discussion

This chapter presents the simulation results obtained for the multilayer spacecraft solar panel under monoenergetic electron irradiation. The transport of incident electrons through the multilayer structure is first analysed in terms of secondary particle production, interaction statistics and penetration behaviour. The resulting energy deposition characteristics are then investigated, followed by the electrostatic quantities derived from the accumulated charge distribution.

4.1 Analysis Display

In the Geant4 setup used, primary electrons undergo multiple scattering and progressively penetrate deeper with increasing incident energy. Secondary gamma particles, generated through electromagnetic interactions, are produced only in small numbers and contribute minimally to the overall transport.



(a) 50 keV (b) 500 keV (c) 5 MeV

Figure 4.1: Representative Geant4 visualization of particle transport through the multilayer spacecraft solar panel for 3 incident electron energies. Electron trajectories are shown in red, while bremsstrahlung photons are shown in blue.

4.2 Secondary Particle Production

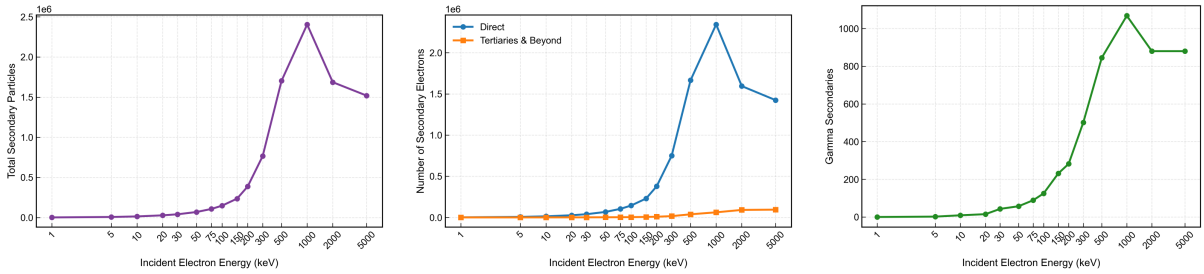
Secondary particles are generated when incident electrons interact with the constituent materials of the solar panel through electromagnetic processes. In the present study, the generated secondary particles consist primarily of electrons produced through ionization and photons produced through bremsstrahlung interactions. Since secondary electrons may remain trapped

inside dielectric layers, their production contributes directly to charge accumulation.

The average secondary particle yield is given by

$$Y = \frac{N_{\text{secondary}}}{N_{\text{incident}}}, \tag{4.1}$$

where $N_{\text{secondary}}$ is the total number of generated secondary particles and N_{incident} is the number of incident primary electrons.



(a) Total (b) Secondaries & Tertiaries (c) Gammas

Figure 4.2: Secondary particle production, with respect to all simulated energies

An increase in the incident electron energy produces a corresponding increase in the total number of secondary particles. The particle trajectories become progressively longer with increasing incident energy, indicating deeper penetration into the multilayer structure before the electrons lose their kinetic energy. Higher-energy electrons undergo a larger number of interactions before either stopping within the structure or exiting the panel, resulting in greater secondary particle generation. Secondary electrons constitute the dominant fraction of the generated particles throughout the investigated energy range, while photon production becomes increasingly noticeable at higher energies due to bremsstrahlung interactions. Since these secondary electrons may subsequently become trapped inside dielectric materials, they form an important source of accumulated charge within the multilayer structure.

4.3 Primary Interaction Statistics

Following entry into the multilayer structure, each incident electron may undergo one of three possible outcomes: transmission through the solar panel, reflection from the surface, or absorption within one of the material layers. The relative proportion of these outcomes depends primarily on the incident electron energy, the dielectric strengths, and stopping capability of the constituent materials.

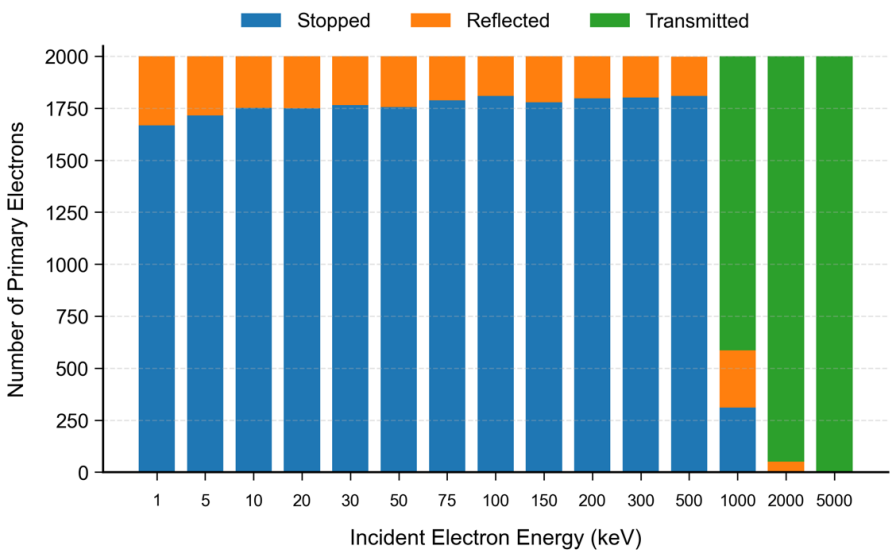


Figure 4.3: Primary interaction statistics for different incident electron energies.

At lower incident energies, most electrons are absorbed within the upper dielectric layers because their kinetic energy is insufficient to penetrate the complete multilayer structure. As the incident energy increases, the transmitted fraction gradually increases, indicating deeper penetration through successive material layers. Reflection is observed primarily for lower-energy electrons due to elastic scattering near the material surface and becomes comparatively less significant as the electron energy increases. The observed transition from absorption-dominated behaviour to transmission-dominated behaviour demonstrates the increasing penetration capability of energetic electrons and directly influences the subsequent energy deposition profile within the solar panel.

4.4 Total Energy Deposition

As incident electrons propagate through the multilayer spacecraft solar panel, a fraction of their kinetic energy is continuously transferred to the surrounding materials through ionization and scattering processes. The total energy deposited within the structure therefore represents the cumulative energy transferred from all incident electrons during their transport.

The deposited energy may be expressed as

$$E_{\text{dep}} = E_{\text{in}} - E_{\text{out}}, \tag{4.2}$$

where E_{in} and E_{out} denote the incident and exiting electron energies, respectively.

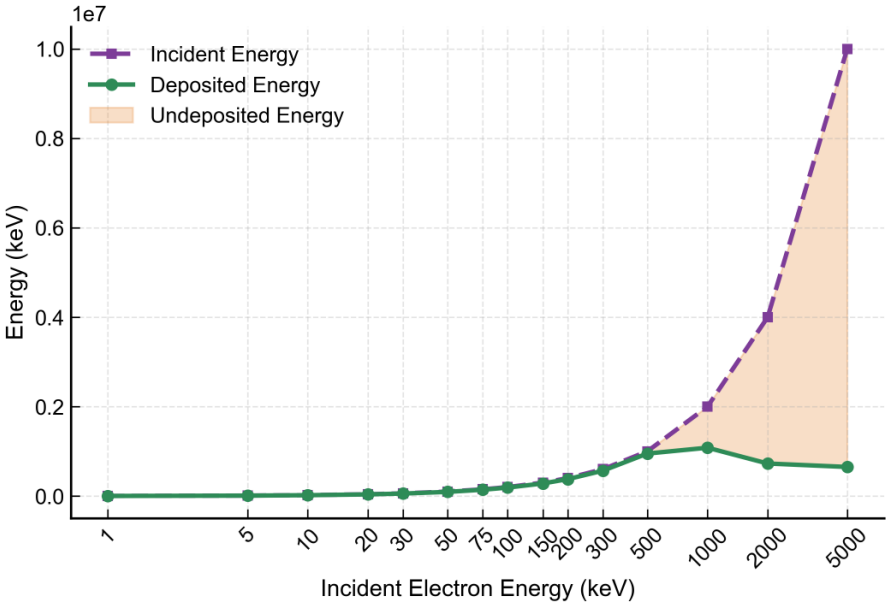


Figure 4.4: Total energy deposited within the multilayer spacecraft solar panel for different incident electron energies.

The deposited energy increases with increasing incident electron energy over the investigated range. Although higher-energy electrons exhibit a greater probability of transmission, their larger initial kinetic energy results in a greater absolute amount of deposited energy within the structure. The deposited energy therefore reflects the combined influence of interaction probability and electron penetration depth.

4.5 Layer-wise Energy Deposition

In addition to the total deposited energy, it is useful to examine how the transferred energy is distributed among the individual layers of the spacecraft solar panel. Owing to differences in density, thickness and atomic composition, each material interacts differently with the incident electron beam, leading to a non-uniform energy deposition profile throughout the multilayer structure.

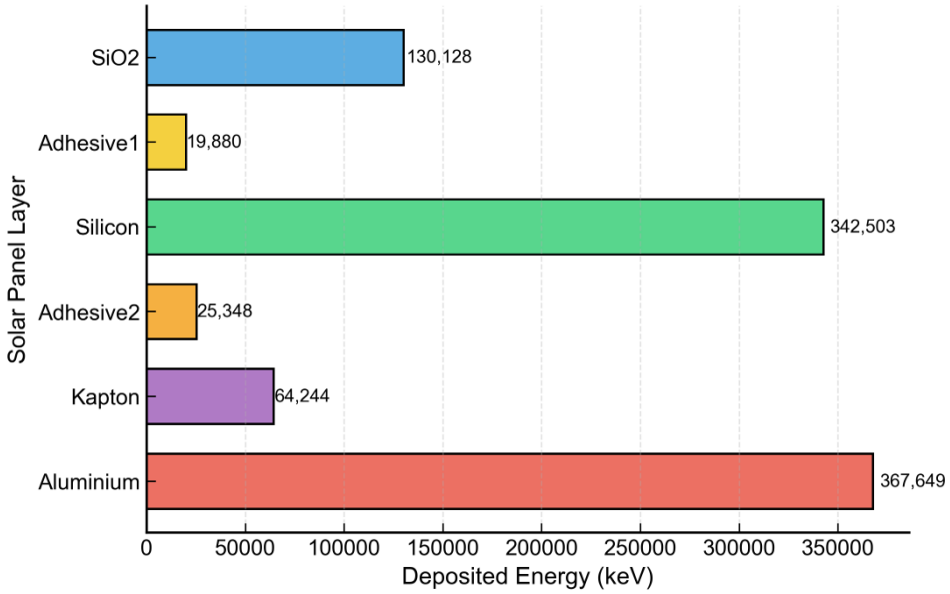


Figure 4.5: Representative layer-wise energy deposition within the multilayer spacecraft solar panel (500 keV energy)

The dielectric cover layers absorb a significant fraction of the energy carried by lower-energy electrons because most particles are unable to penetrate beyond the upper portion of the structure. As the incident energy increases, electrons travel deeper into the multilayer assembly, resulting in greater energy deposition within the silicon solar cell and the aluminium substrate. The observed distribution reflects the progressive transport of energetic electrons across successive material interfaces and demonstrates that deposited energy is strongly influenced by both electron energy and layer properties.

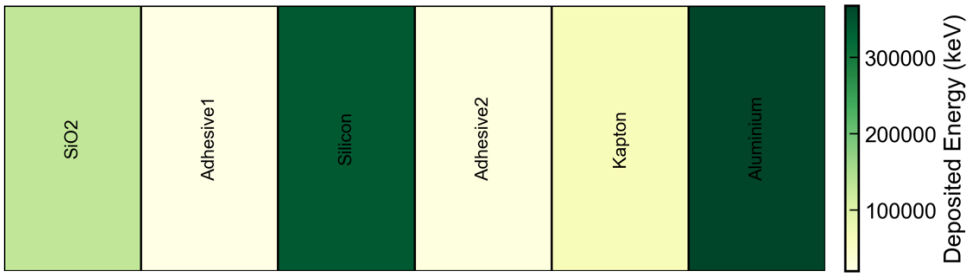


Figure 4.6: Comparative layer-wise energy deposition within the multilayer spacecraft solar panel (500 keV energy)

The representation above highlights the deposited energy within thinner dielectric layers that is less apparent on the linear scale. Although aluminium dominates the total deposited energy at higher incident energies, measurable energy deposition is observed throughout the multilayer structure, demonstrating continuous energy transfer across all constituent materials.

4.6 Depth Dose Profile

The depth-dose profile describes the variation of deposited energy as a function of penetration depth within the multilayer solar panel. Unlike the layer-wise energy deposition, which provides integrated values for individual materials, the depth-dose profile illustrates the continuous spatial distribution of energy transferred by the incident electrons throughout the structure.

The absorbed dose is defined as

$$D = \frac{E_{\text{dep}}}{m}, \tag{4.3}$$

where E_{dep} is the deposited energy and m is the mass of the irradiated material.

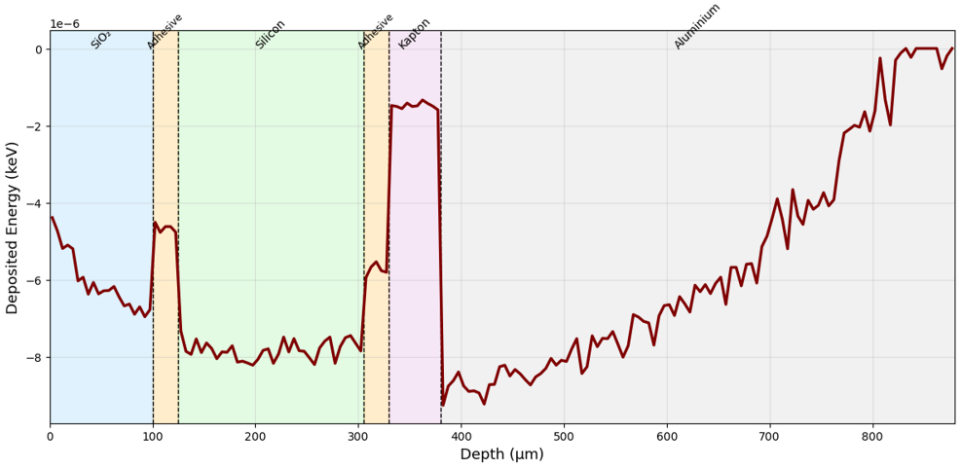


Figure 4.7: Depth-dose profile across the multilayer spacecraft solar panel (500 keV energy)

The deposited dose is concentrated near the surface for lower-energy electrons because their penetration depth is limited by rapid energy loss within the dielectric layers. As the incident energy increases, the location of maximum deposited dose shifts progressively deeper into the multilayer structure. This behaviour demonstrates that the spatial distribution of absorbed energy depends strongly on the incident electron energy. Since charge accumulation originates from electrons that lose their energy within the dielectric materials, the depth-dose profile provides a direct connection between electron transport and the electrostatic behaviour analysed in the following sections.

4.7 Charge Accumulation and Charge Density

As energetic electrons lose their kinetic energy within the dielectric layers of the spacecraft solar panel, a fraction of the incident particles remains trapped inside. These trapped electrons result in the development of negative charge, which is the basis of deep dielectric charging. Accumulated charge is determined from the total number of trapped electrons as

$$Q = ne, \tag{4.4}$$

where n is the number of accumulated electrons and e is the elementary charge.

Since the individual layers possess different dimensions, the accumulated charge alone does not adequately represent the severity of charging. A more meaningful parameter is the charge density, defined as

$$\rho = \frac{Q}{V}, \tag{4.5}$$

where V is the volume of the corresponding material layer.

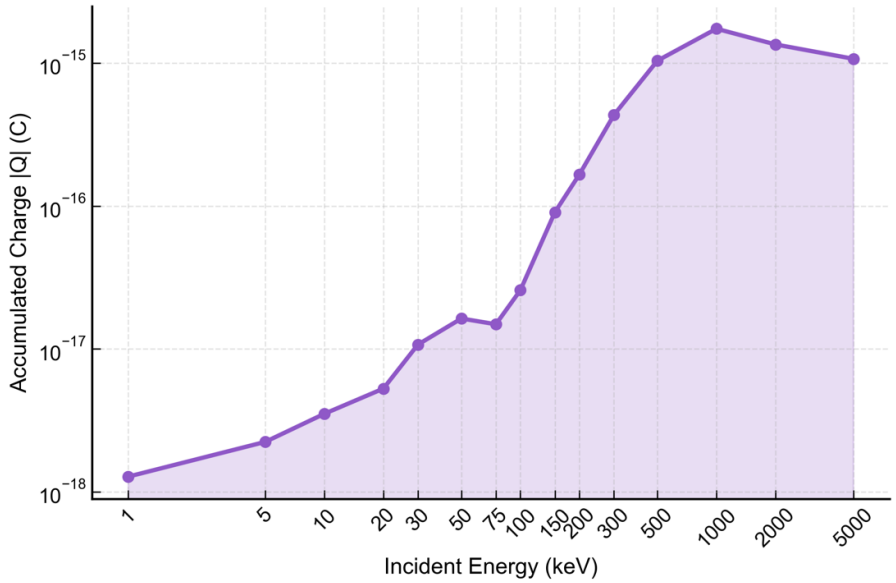


Figure 4.8: Accumulated charge within the individual layers of the multilayer spacecraft solar panel.

The accumulated charge is negative because electrons constitute the trapped species. It varies considerably among the constituent layers owing to differences in electron penetration depth, stopping behaviour and material properties. While thicker layers may accumulate a larger total charge, the corresponding charge density depends on both the deposited charge and the material volume. Consequently, layers exhibiting the highest charge density represent regions where electrostatic effects are expected to be most significant. These localized charge concentrations subsequently generate internal electric fields within the multilayer structure.

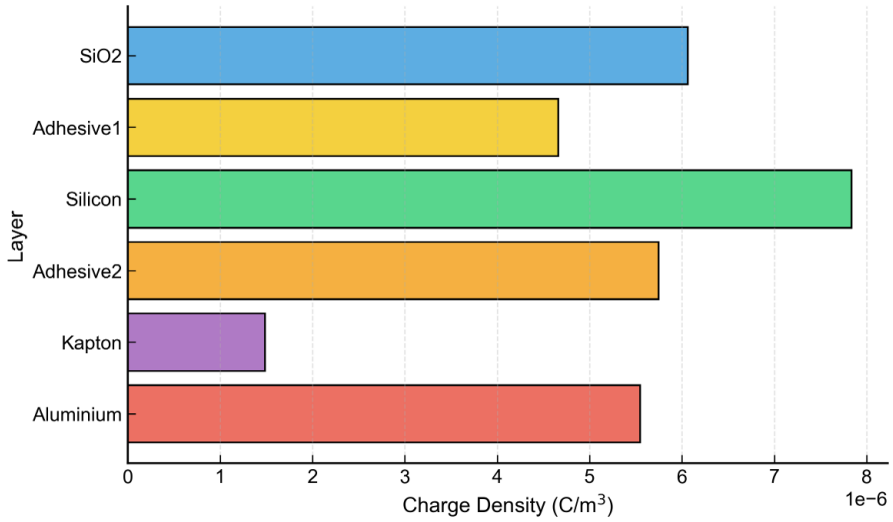


Figure 4.9: Comparative representation of Charge density within the individual layers of the multilayer spacecraft solar panel (500 keV energy)

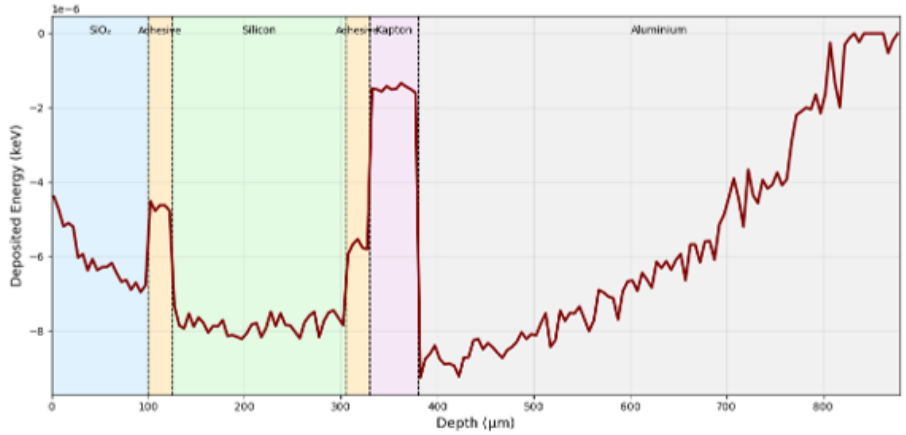


Figure 4.10: Charge density profile across the multilayer spacecraft solar panel (500 keV energy)

The variation in charge density can be thus illustrated over several orders of magnitude, making the response of individual material layers easier to compare. The dielectric layers exhibit noticeable charge accumulation due to electron trapping, whereas the aluminium substrate shows comparatively lower charge density because of its conductive nature. The distribution demonstrates that the accumulated charge is not uniform throughout the structure but depends strongly on both the incident electron energy and the material properties of each layer.

4.8 Electric Field Estimation

The non-uniform distribution of accumulated charge produces an internal electric field within the multilayer spacecraft solar panel. This electric field originates from spatial variations in the electrostatic potential and governs the forces experienced by charged particles trapped inside the dielectric materials. The electric field is related to the electric potential through

$$\mathbf{E} = -\nabla V, \tag{4.6}$$

where $\mathbf{E}$ denotes the electric field vector and V represents the electric potential.

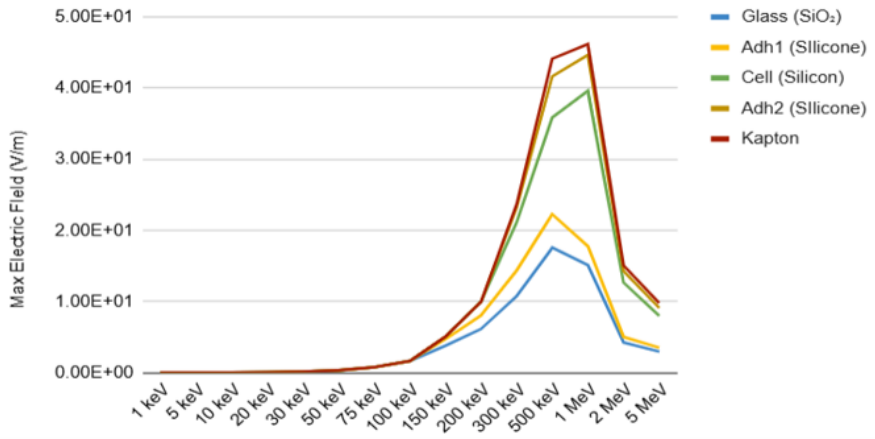


Figure 4.11: Estimated electric field produced by the accumulated charge in different materials within the multilayer spacecraft solar panel.

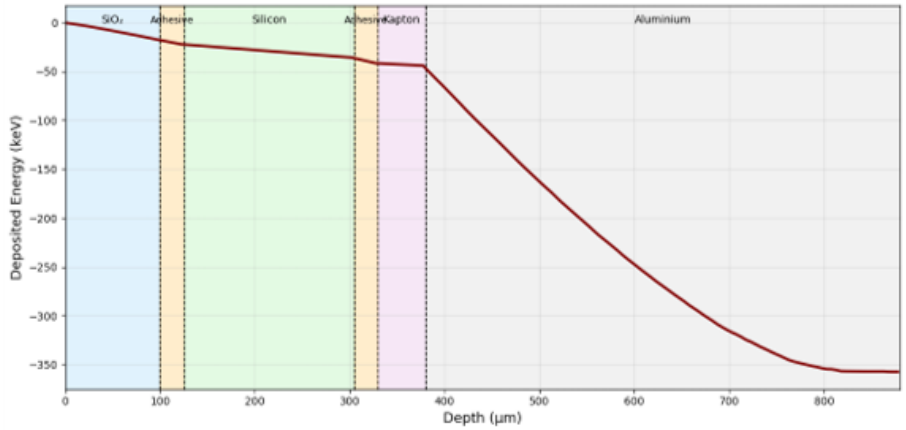


Figure 4.12: Electric field profile across the multilayer spacecraft solar panel (500 keV energy)

The electric field strength depends directly on the magnitude and spatial distribution of accumulated charge within the dielectric layers. Regions containing higher localized charge densities generate correspondingly stronger electric fields, resulting in larger electrostatic stresses inside the material. The calculated electric field therefore provides an indication of those regions that are comparitively most susceptible to deep dielectric charging effects.

4.9 Potential Difference

The electric field generated by the accumulated charge establishes a potential difference across the multilayer spacecraft solar panel. The potential difference between two points is obtained by integrating the electric field along the direction normal to the material layers,

$$\Delta V = -\int \mathbf{E} \cdot d\mathbf{l}, \tag{4.7}$$

where $d\mathbf{l}$ denotes the differential path element along the integration direction.

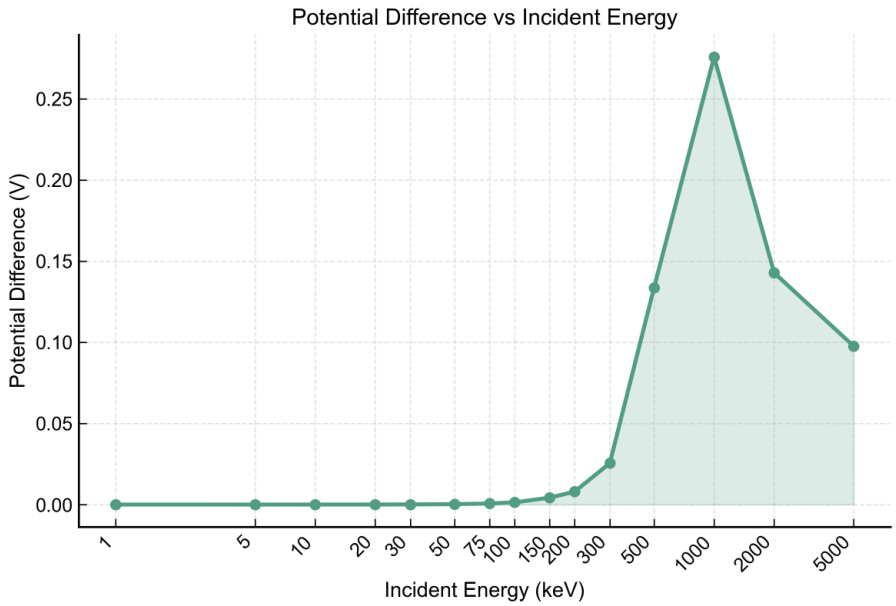


Figure 4.13: Potential difference developed across the multilayer spacecraft solar panel due to accumulated charge.

The potential difference represents the cumulative effect of the electric field across the complete multilayer structure rather than within an individual layer. The calculated potential difference increases with the magnitude of accumulated charge within the dielectric layers and represents the overall electrostatic state of the multilayer structure. Since the electric field is directly related to the potential gradient, larger charge concentrations lead to larger potential differences across the material stack. The combined analysis of charge accumulation, charge density, electric field and potential difference provides a comprehensive description of the electrostatic response of the multilayer spacecraft solar panel under electron irradiation.

Conclusion

This internship focused on the development of a Geant4-based simulation framework for investigating deep dielectric charging in multilayer spacecraft solar panels subjected to energetic electron irradiation. A simplified multilayer geometry consisting of SiO_2, silicone adhesive, silicon, Kapton and aluminium was implemented to represent the essential structural components of a spacecraft solar array. The developed framework combines Monte Carlo particle transport simulations with ROOT-based post-processing routines to analyse electron transport, energy deposition and the resulting electrostatic behaviour within the multilayer structure.

The framework enables a systematic investigation of the sequence of physical processes leading to deep dielectric charging. Beginning with the interaction of incident electrons with the multilayer geometry, the simulation records secondary particle production, primary interaction statistics and energy deposition. The generated simulation data are subsequently processed to determine charge accumulation, charge density, electric field and potential difference, thereby establishing a continuous workflow from microscopic particle interactions to macroscopic electrostatic quantities.

5.1 Major Findings

The simulation results demonstrate that the transport behaviour of energetic electrons is strongly dependent on the incident electron energy. As the incident energy increases, the number of generated secondary particles increases due to the greater number of electromagnetic interactions occurring within the multilayer structure. Simultaneously, the interaction statistics exhibit a gradual transition from absorption-dominated behaviour at lower energies to transmission-dominated behaviour at higher energies, reflecting the increased penetration capability of energetic electrons.

The energy deposition analysis shows that low-energy electrons deposit most of their energy within the upper dielectric layers, whereas higher-energy electrons penetrate progressively deeper into the solar panel before depositing energy throughout the multilayer stack. Consequently, both the total energy deposition and the depth-dependent energy deposition profiles demonstrate that increasing incident energy shifts the region of maximum energy deposition further into the structure, producing a non-uniform distribution of deposited energy across the individual material layers.

The electrostatic analysis demonstrates that trapped electrons accumulate within the dielectric materials, producing measurable charge accumulation and charge density throughout the multilayer structure. The resulting non-uniform charge distribution generates internal electric fields and corresponding potential differences across the solar panel. These electrostatic quantities provide useful indicators for assessing the susceptibility of dielectric materials to deep dielectric charging under energetic electron irradiation.

Overall, the work carried out during this internship establishes a foundation for future investigations of deep dielectric charging using Geant4 and demonstrates the feasibility of extending Monte Carlo particle transport simulations towards spacecraft charging applications.

5.2 Limitations and Future Scope

The present work represents an initial framework for investigating deep dielectric charging in multilayer spacecraft solar panels. Several simplifying assumptions were adopted to maintain computational efficiency, including the use of monoenergetic electron beams, a simplified multilayer geometry and static charge accumulation analysis. Consequently, the present framework does not account for the temporal evolution of accumulated charge or realistic space radiation spectra.

Future developments may extend the framework by incorporating time-dependent charge transport models, realistic trapped-particle energy spectra and self-consistent electrostatic field solvers. Additional improvements may include more detailed spacecraft geometries, experimentally validated material properties, and comparisons with analytical or experimental charging data. Such extensions would improve the predictive capability of the framework and enable more realistic assessments of deep dielectric charging in operational spacecraft environments.

Overall, the developed simulation framework provides a flexible foundation for studying electron transport and electrostatic charging phenomena in spacecraft dielectric materials. The modular architecture adopted during this internship allows additional physical models and analysis techniques to be incorporated with minimal modifications, making the framework suitable for future investigations into spacecraft charging and radiation effects.

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